How-To Library

Informatica

Address Verification Best Practices for Australia

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Abstract

This document tells you how to connect the elements in an Australia address to the appropriate address elements in Informatica® Address Verification. This document also describes best practices that you can implement when you verify addresses in Australia.

Supported Versions

- Informatica Address Verification 3.0 and later
- Informatica Address Verification (On-Premises) 5.x

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Introduction

Informatica Address Verification can validate and enhance addresses in Australia. This how-to article provides you with configuration information and best practices that you can follow to get the most from the addresses. The best practices include advice on how to select the appropriate XML elements for Australia address data.

To verify the quality of an input address, Address Verification compares the address information to the data in one or more reference data files. You can configure Address Verification to identify the different types of address information in the input data.

When you select the input address elements, you must ensure that the elements are correct for the different types of input information. The country that the address identifies determines the elements that you need.

You must map each item of address information to an appropriate element in Address Verification.

Process Modes

Informatica Address Verification supports the following process modes for Australia addresses:

- BATCH
- CERTIFIED

- COUNTRY RECOGNITION
- FAST COMPLETION
- INTERACTIVE
- PARSE

Character Set

Australia reference address databases store address data in the Latin-1 script. By default, Informatica Address Verification returns the verified Australia addresses in the Latin-1 script.

Certified Mode Validation of Australia Addresses

You can use certified mode to verify Australia addresses in accordance with the Address Matching Approval System (AMAS) guidelines of Australia Post. To verify Australia addresses in certified mode, you must install the `AUS5C1.MD` and `AUS5C2.MD` databases and include the corresponding unlock codes in `SetConfig.xml`.

The `AUS5C1.MD` and `AUS5C2.MD` databases contain Postal Address File (PAF) data, which includes Australia Post's Delivery Point Identifiers (DPIDs).

To enable certified mode verification of Australia addresses, set the `EnrichmentAMAS` attribute to ON.

Address Format and Address Structure

You improve the performance of Address Verification when you define the input address in the format that the postal service expects.

The Australia postal service recognizes the following structure for a postal address:

```
Line 1: RECIPIENT OR CONTACT NAME
Line 2: [SUB-BUILDING INFORMATION] HOUSE NUMBER STREET NAME STREET TYPE
Line 3: CITY PROVINCE POSTAL CODE
Line 4: COUNTRY
```

The following sample address illustrates the structure:

```
SHOP 7 208 ADELAIDE ST
BRISBANE CITY  QLD  4000
AUS
```

Key Address Elements

Address Verification is unlikely to verify an address successfully if one of the following items is absent from the address:

- Locality
- Postal Code
- Street

Mapping Input Address Data to Address Elements

To verify an address, map the input address information to appropriate elements in Address Verification.

The following table shows how you can map the input address to the address elements:

Information Type	Address Information	Address Elements
Street Name, Street Type	ADELAIDE ST	Street
House Number	208	HouseNumber
Apartment, Floor	SHOP 7	SubBuilding
City	BRISBANE CITY	Locality
State	QLD	ProvinceAbbreviation
Postal Code	4000	PostalCode

Input and Result XML in Address Verification (On-Premises)

The following example shows how you can map the address information to the XML address elements in Address Verification (On-Premises).

You enter the following input address:

```
<InputData>
    <AddressElements>
        <Country Item="1" Type="NAME">AUS</Country>
        <Locality Item="1" Type="COMPLETE">BRISBANE CITY</Locality>
        <PostalCode Item="1" Type="UNFORMATTED">4000</PostalCode>
        <Province Item="1" Type="COUNTRY_STANDARD">QLD</Province>
        <Street Item="1" Type="COMPLETE">ADELAIDE ST</Street>
        <Number Item="1" Type="COMPLETE">208</Number>
        <SubBuilding Item="1" Type="COMPLETE">SHOP 7</SubBuilding>
    </AddressElements>
</InputData>
```

Informatica Address Verification (On-Premises) returns the following result:

```
<?xml version="1.0" encoding="UTF-16"?>

<!-- EngineVersion="5.13.1.45876" -->

<Result     ProcessStatus="V4"
     ModeUsed="BATCH"
     Count="1"
     CountOverflow="NO"
     CountryISO3="AUS"
     PreferredScript="LATIN_1"
     PreferredLanguage="ENGLISH">

<ResultData ResultNumber="1"
     MailabilityScore="5"
     ResultPercentage="100.00"
     ElementResultStatus="F0F0F0F0F00000F000E0"
     ElementInputStatus="60606060600000600060"
     AddressResolutionCode="00000000000000000000"
     ExtElementStatus="00000000000000000000"
     ElementRelevance="10101010100000100010"
     AddressType="S"
     LanguageISO3="ENG">
    <AddressElements>
        <Country Type="NAME_EN" Item="1">Australia</Country>
```

```
        <Locality Item="1">Brisbane City</Locality>
        <PostalCode Item="1">4000</PostalCode>
        <Province Item="1">QLD</Province>
        <Street Item="1">Adelaide St</Street>
        <Number Item="1">208</Number>
        <SubBuilding Item="1">Shop 7</SubBuilding>
    </AddressElements>
    <AddressLines>
        <DeliveryAddressLine Line="1">Shop 7 208 Adelaide St</DeliveryAddressLine>
        <CountrySpecificLocalityLine Line="1">Brisbane City  QLD  4000</
CountrySpecificLocalityLine>
        <FormattedAddressLine Line="1">Shop 7 208 Adelaide St</FormattedAddressLine>
        <FormattedAddressLine Line="2">Brisbane City  QLD  4000</FormattedAddressLine>
        <FormattedAddressLine Line="3">Australia</FormattedAddressLine>
    </AddressLines>
    <AddressComplete>Shop 7 208 Adelaide St
Brisbane City  QLD  4000
Australia</AddressComplete>
</ResultData>
</Result>
```

Input and Result XML in Address Verification

The following example shows how you can verify an Australia address in a SOAP call to Address Verification.

You submit the following request:

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope" xmlns:v="http://
validator5.AddressDoctor.com/Webservice5/v4">
 <soap:Header/>
 <soap:Body>
    <v:Process>
      <v:login>****</v:login>
      <v:password>*****</v:password>
      <v:parameters>
         <v:ProcessMode>BATCH</v:ProcessMode>
         <v:ServiceParameters></v:ServiceParameters>
         <v:ValidationParameters>
         </v:ValidationParameters>
      </v:parameters>
      <v:addresses>
       <v:Address>
        <v:FormattedAddress>
           <v:string>SHOP 7/208 ADELAIDE ST</v:string>
           <v:string>BRISBANE CITY QLD 4000</v:string>
           <v:string>AUS</v:string>
        </v:FormattedAddress>
       </v:Address>
      </v:addresses>
      <v:enrichments>
      </v:enrichments>
    </v:Process>
 </soap:Body>
</soap:Envelope>
```

Informatica Address Verification returns the following response:

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema">
   <soap:Body>
      <ProcessResponse xmlns="http://validator5.AddressDoctor.com/Webservice5/v4">
         <ProcessResult>
            <StatusCode>100</StatusCode>
            <StatusMessage>OK</StatusMessage>
            <Results>
               <Result>
                  <ProcessStatus>V4</ProcessStatus>
                  <CountryISO3>AUS</CountryISO3>
                  <ResultDataSet>
                     <ResultData>
```

```
                        <ResultNumber>1</ResultNumber>
                       <MailabilityScore>5</MailabilityScore>
                        <ResultPercentage>100.00</ResultPercentage>
                        <ElementInputStatus>6060606060000600060</ElementInputStatus>
                        <ElementResultStatus>F0F0F0F0F00000F000E0</ElementResultStatus>
                        <ElementRelevance>1010101010000100010</ElementRelevance>
                        <ExtElementStatus>00000000000000000000</ExtElementStatus>
                        <AddressResolutionCode>000000000000000000000</AddressResolutionCode>
                        <AddressType>S</AddressType>
                        <LanguageISO3>ENG</LanguageISO3>
                        <Address>
                           <SubBuilding>
                              <string>SHOP 7</string>
                           </SubBuilding>
                           <Street>
                              <string>ADELAIDE ST</string>
                           </Street>
                           <HouseNumber>
                              <string>208</string>
                           </HouseNumber>
                           <Locality>
                              <string>BRISBANE CITY</string>
                           </Locality>
                           <PostalCode>
                              <string>4000</string>
                           </PostalCode>
                           <Province>
                              <string>QLD</string>
                           </Province>
                           <Country>
                              <string>AUSTRALIA</string>
                           </Country>
                           <DeliveryAddressLines>
                              <string>SHOP 7 208 ADELAIDE ST</string>
                           </DeliveryAddressLines>
                           <CountrySpecificLocalityLine>
                              <string>BRISBANE CITY  QLD  4000</string>
                           </CountrySpecificLocalityLine>
                           <FormattedAddress>
                              <string>SHOP 7 208 ADELAIDE ST</string>
                              <string>BRISBANE CITY  QLD  4000</string>
                              <string>AUSTRALIA</string>
                           </FormattedAddress>
                           <AddressComplete>SHOP 7 208 ADELAIDE ST;BRISBANE CITY  QLD
4000;AUSTRALIA</AddressComplete>
                        </Address>
                     </ResultData>
                  </ResultDataSet>
               </Result>
            </Results>
         </ProcessResult>
      </ProcessResponse>
   </soap:Body>
</soap:Envelope>
```

Address Enrichments

You can configure Informatica Address Verification to add enrichments to verified Australia addresses.

The following table lists the types of enrichment that you can add to an Australia address:

Enrichment Type	Description
CAMEO	You can retrieve CAMEO profiles as enrichments for valid Australia addresses. CAMEO profiles are consumer classification systems that provide socio-economic and demographic indicators about the residents at the address.
Standard Geocoding	You can retrieve standard, interpolated geocoordinates for valid Australia addresses.
Arrival Point Geocoding	You can retrieve arrival point geocoordinates for valid Australia addresses.

Frequently Asked Questions

Does Address Verification use Geocoded National Address File (G-NAF) data?

No. Address Verification does not use G-NAF data.

Why does Address Verification reject an address in certified mode when the street information falls on the border of two localities?

AMAS certification defines Bordering Locality Matching (BLM) rules for addresses on the border between two localities. To comply with the rules, Address Verification does not change an address that falls on the border of two localities in certified mode. Address Verification returns a process status code of Ix for the address and copies the input to the output elements.

For example, Address Verification returns an I4 status code for the following address because the address is located on the border of the Ramsgate Beach and Sans Souci localities:

```
138 Chuter Ave
Ramsgate Beach NSW 2217
AUS
```

Does Address Verification use Incremental Change File (ICF) data?

No, but you can download the file from Informatica and consult it to find addresses for which the DPID value has recently changed.

Does Address Verification return DPID values in batch mode?

No, Address Verification returns DPID values in certified mode only.

What are the criteria for address certification to the AMAS standard?

Australia Post maintains a set of stringent criteria for the certification of any address to the AMAS standard. Australia Post publishes a handbook that describes the criteria. For complete information on the certification criteria, consult the AMAS handbook.

How does batch mode differ from certified mode?

The AMAS criteria can cause Address Verification to reject an address in certified mode that could produce a reference data match in batch mode. For example, the AMAS criteria state that an address verification process must reject a phantom primary point address.

A phantom primary point is an address that identifies a verifiable street destination but not a mailbox for delivery. That is, the address provides primary information, or information to street and building number level, but it does not provide required secondary information, such as a sub-building number. The street address of an apartment building is a phantom primary point if the building itself does not receive mail.

Note: You can add the Address Resolution Code field to the output address to identify missing elements in the address. An Address Resolution Code value of 2 in any position in the address structure indicates that a data element is absent but required at that position.

Reference Address Databases

Informatica updates the Australia reference address databases monthly. You can download the latest database packages from the Informatica Database Download portal. For news about reference address databases and software updates for Address Verification, subscribe to the newsletter from Informatica.

Informatica Address Verification provides the following reference databases for Australia addresses:

- `AUS5BI.MD`. Batch and interactive modes.
- `AUS5C1.MD` and `AUS5C2.MD`. Certified mode.
- `AUS5CA.MD`. CAMEO data.
- `AUS5FC.MD`. Fast completion mode.
- `AUS5GC.MD`. Standard geocoding.
- `AUS5GCAP.MD`. Arrival point geocoding.

You download databases for Informatica Address Verification (On-Premises). You do not need to download databases for Informatica Address Verification, as Informatica Address Verification runs on Informatica Cloud®.

Note: The reference data zip files that you can download for Australia can include the two-digit codes _01 and _02 in the file names. Select the _02 version of the file to download reference data that includes sub-building data. Select the _01 version of the file to download reference data without sub-building data.

Additional Documentation Resources

To view the latest documentation for Informatica Address Verification, click the following links:

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor-cloud

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